

1 Classifications of Econometric Models

This note is inspired by Xiaohong Chen's 2023 guest lecture at the University of Chicago. The key difference in the following four models is in the *dimensionality of the parameters of interest*.

An econometric model is a family of probability distributions (of observed random variables) indexed by unknown parameters.

A model is called:

1. ***Parametric***, if all of its parameters are in finite-dimensional parameter spaces.

Example:

- Standard OLS regression to fit a linear model.

2. ***Nonparametric***, if all of its parameters are in infinite-dimensional parameter spaces.

Some examples:

- Empirical CDF.
- Kernel Density Estimation. Note that even though we have to specify a bandwidth, it is not a parameter of the DGP or model, but rather a tuning parameter chosen for the estimation algorithm.

3. ***Semiparametric***, if its parameters of interest are in finite-dimensional spaces but its nuisance parameters¹ are in infinite-dimensional spaces.

Some examples:

- GMM (Hansen, 1982). Consider the moment restriction

$$\mathbb{E}[g(Z, \theta_0)] = \int g(z, \theta_0) dF_0(z) = 0.$$

The parameter of interest is $\theta_0 \in \mathbb{R}^p$, which is finite dimensional. However, the true joint distribution $F_0 \in \mathcal{F}$ lies in an infinite-dimensional space.

¹A nuisance parameter is a parameter of the DGP that is not of primary interest, but is necessary to fully specify the distribution. For example, consider a linear regression model with $Y = X'\beta + \epsilon$ where $\epsilon \sim N(0, \sigma^2)$. If the target is β , then σ^2 is a finite-dimensional nuisance parameter.

- Partially Linear Regression (Robinson, 1988). Consider the model

$$Y = X'\beta_0 + g_0(Z) + \epsilon, \quad \mathbb{E}[\epsilon|X, Z] = 0.$$

The parameter of interest is β_0 , which lies in a finite dimensional space.

The nuisance parameter are g_0 and the conditional distribution $F_{\epsilon|x,z}$.

4. ***Semi-nonparametric***, if it contains both finite-dimensional and infinite-dimensional unknown parameters of interest.

Some examples:

- In the identification of ascending auctions, Ponomarev and Wen (2026) proposed a method to bound the distribution of highest valuations under certain dependence assumptions, and then use those bounds to construct bounds on the profit function. The model is semi-nonparametric, because the correlation parameter $\rho \in [0,1]$ lies in a finite dimensional space, while the empirical CDF lies in an infinite dimensional space. Both these parameters are of interest since they are used to construct the profit bound functionals.
- Consider a conditional moment restriction model given by

$$\mathbb{E}[\rho(Y, X; \theta_0, g_0(Z)) | W] = 0$$

where ρ is the moment function, θ_0 is a finite-dimensional parameter of interest, g_0 is an infinite-dimensional parameter of interest, and W are some conditioning instruments which may overlap with X, Z . Examples include:

- IV regression specified as

$$Y = X'\theta_0 + g_0(Z) + \varepsilon, \quad \mathbb{E}[\varepsilon | W] = 0$$

so that $\rho(Y, X; \theta, g) = Y - X'\theta - g(Z)$.

- Demand for J differentiated products across each market t , specified as

$$P_{jt} = \sigma_j(S_t, X_t) + W'_{jt}\gamma_0 + \xi_{jt}, \quad j = 1, \dots, J$$

where P_{jt} is the price, S_t is the vector of market shares, X_t are the observed product characteristics, W_{jt} are observed cost shifters, and ξ_{jt} is the unobserved demand shock such that $\mathbb{E}[\xi_{jt} \mid Z_t] = 0$. We then have

$$\mathbb{E} [P_{jt} - W'_{jt}\gamma_0 - \sigma_j(S_t, X_t) \mid Z_t] = 0$$

so that the conditional moment restriction is $\rho_j(P_t, S_t, X_t, W_t; \gamma, \sigma_j) = P_{jt} - W'_{jt}\gamma - \sigma_j(S_t, X_t)$.

Here, the finite dimensional parameter of interest is γ_0 (e.g. as a cost-shifter), while the infinite-dimensional parameter of interest is the nonparametric inverse demand function σ_j .